
Predicting the Next Word in Text Messages Using Machine Learning

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1 Research Question and Motivation

1.1 Clearly state the research question you aim to answer

Can AI-powered machine learning models accurately predict the next word in text messages using historical user data, and how can such predictions improve personalization in messaging apps and improve typing efficiency?

1.2 Provide context and motivation for why this question is important or interesting.

Predicting the next words improves users' texting efficiency, and reduces repetitive typing on mobile devices. Personalized recommendations, tailored to a user's writing style and vocabulary, make messaging applications more engaging. This project would demonstrate use of natural language processing (NLP) in messaging apps and how AI can learn patterns in language usage by analyzing historical user data. Additionally, machine learning models can adapt over time to changing user habits and preferences, providing more accurate and personalized predictions.

1.3 Discuss any relevant background knowledge or prior work.

In the Data Management (198:210) course, the final project was similar to the requirements of this project. A group project focused on analyzing kaggle dataset and applying data processing and analytical techniques to extract meaningful insights. In that project, our team worked with structured data using Python and libraries such as pandas, matplotlib, and scikit-learn. The project involved several stages (Data Science Cycle), including data cleaning, preprocessing, exploratory data analysis (EDA), and building machine learning models to analyze patterns in the data. Similar to the current project, the goal was to understand important features within a dataset and use statistical or machine learning approaches to support predictions and data-driven conclusions.

2 Data Sources

2.1 Identify the data sources you plan to use.

For this project, we plan to use four text datasets from Kaggle. These datasets provide a mix of conversational text (ie, email, SMS messages, chatbot history, daily dialogs), which we'll use to train our next-world prediction model. Here is the detail, along with a link to the datasets.

- DailyDialog Dataset (Kaggle) - This dataset contains dialogues with emotions. We will use these two columns to get natural conversation text.
- Multi-Turn Chatbot Conversation Dataset (Kaggle) - This dataset has quite a few conversational examples suitable for learning conversational patterns.

- SMS Spam Collection Dataset (Kaggle) - This dataset contains real SMS messages. We will use the SMS column and ignore the spam/ham labels, since we only need the message content.
- Enron Email Dataset (Kaggle) - This dataset contains real email conversations. We will use the message column and drop the extra column, such as date, from, to, etc. This set will provide a professional conversation format and longer-form text examples.

2.2 Explain how you will obtain, clean, and preprocess the data.

All datasets will be downloaded directly from Kaggle in CSV if the dataset is in CSV format; for others, we will either download a .txt file or use the **kagglehub** library and API call. For text cleaning and preprocessing, we will convert all text to lowercase, remove punctuation, emojis, and special characters, remove stop words to reduce very common, less informative words, tokenize sentences into words, and create hard-coded sequences where the model can learn to predict the next word. As a result, the model has access to clean, well-structured text and can learn patterns.

2.3 If applicable, discuss potential challenges in data access, quality, or preprocessing.

Potential challenges in data access is data privacy, especially with the email and messages dataset. For quality and preprocessing, inconsistent text, such as typos, abbreviations, slang, and different personal formatting in the message. Also, it would be unbalanced word frequency, some words may appear very often while others are rare, which can affect model learning. However, preprocessing and cleaning will help reduce quality issues and make the datasets usable to generate a model to predict the next word.

3 Methodology and Analysis Plan

3.1 Describe the data science techniques you intend to use.

We will use Natural Language Processing (NLP) and ML techniques to predict the next word in text input. Idea of a project to train the model to recognize the sequence patterns and predict the next word. To do so on time and correctly will follow the data science cycle.

3.2 Outline your approach for data exploration, modeling, and evaluation.

Starting with the first step, data collection, then preprocessing and cleaning as mentioned in the data sources section, and then to data exploration, we plan to analyze/extract by checking sentence lengths, the number of messages across all four sets in total, and identify common words, etc. With all this info and cleaning, we will proceed with the EDA graph and analysis patterns, and charts. Then, for modeling, we're not sure at this point, but after research, we may use N-gram models to start predicting the next word based on the previous few words (we are still exploring more options). Lastly, for evaluation, use precision, recall, F1-score, and/or linear regression to measure the accuracy of predictions.

3.3 Specify the tools and libraries (e.g., Python, R, Pandas, Scikit-Learn, TensorFlow) you plan to utilize.

We will use Python and its libraries, such as pandas and Numpy, for data processing and cleaning, NLTK for NLP, Scikit-learn, and TensorFlow for machine learning, and Matplotlib and similar for visualization.

4 Expected Outcomes and Evaluation

4.1 Discuss the anticipated findings or insights you hope to uncover.

This project anticipates that machine learning models will be able to predict the next word in text messages with relatively high accuracy using historical user data, while also identifying which models

perform best for chat based messaging. It is expected to provide insight into how individual writing habits influence the performance of prediction models. It will also examine whether factors such as text length, word sequences, and punctuation improve the accuracy of predictions. Additionally, the project aims to demonstrate how next word prediction can enhance the user's experience in messaging applications by reducing typing effort and minimizing errors. Finally, the findings may offer insight into how messaging apps and keyboards can integrate predictive text models to improve usability and user retention.

4.2 Define metrics or evaluation criteria for assessing your results.

The next word prediction model's performance will be evaluated using several metrics. Precision measures how many of the words predicted by the model are actually correct, indicating the accuracy of the model's suggestions. Recall measures how many of the correct next words in the dataset are successfully identified by model. The F1 score combines both precision and recall to provide balanced evaluation of the model's performance. In addition, perplexity will be used as an evaluation metric to measure how well the model predicts sequence of words in NLP. Lower perplexity indicates better predictive performance. Together these metrics and evaluation criteria will help determine which model performs best for text predicting the next word in messaging applications.

4.3 If applicable, outline any potential extensions or broader implications of your work.

Potential extensions for the next word prediction model include expanding it to support multiple languages and users who communicate in more than one language. The model could also be enhanced to improve accessibility for users with disabilities by reducing the amount of typing. Additionally, predictive text features could be integrated into various platforms such as search engines, email services, and messaging applications. Insights from this project may also contribute to a better understanding of how AI generates suggestions and responses in text based applications. Furthermore, the model could also demonstrate how AI adapts to individual user preferences, which can enhance user experience and encourage continued use of predictive features.